INFORMATION SECURITY (IS)

Info types: Personal, business, financial, intellectual, system
Components of an Information System (IS): Software, Hardware, Data, People, Procedures, Networks
Bottom-Up: Implementations happen before policies
Top-Down: Initiated by management, more effective
McCumber cube: x: CIA (what), y: IS (where), z: Ctrl (how)
(C)onfidentiality: Prevent unauthorized access to info
(I)ntegrity: Protect reliability and correctness of information
(A)vailability: Ensure uninterrupted access to information
Information states: In **storage**, in **transit**, in **use**
Control: A measure designed to reduce the potential risk of an attack by eg. **Policy, Education, Technology**
(S)poofing: Pretending to be someone else
(T)ampering: Unauthorized data modification or altering
(R)epudiation: Denying actions without proof
(I)nformation disclosure: Exposing sensitive information
(D)enial of service: Making systems or services unavailable
(E)levation of privilege: Gaining unauthorized rights

THREAT CATEGORIZATION

SOCIAL ENGINEERING

Manipulating people to reveal confidential information
Spear Phishing: Targeted phishing to specific individuals
Vishing: Voice-based phishing over phone or video calls
Smishing: SMS / Text-based phishing

SOFTWARE ATTACKS

Exploiting vulnerabilities in software to gain access/steal data
Virus: Malware that attaches to programs and spreads
Worms: Self-replicating malware that spreads over a network
Trojan Horse: Malicious SW disguised as legitimate apps
Ransomware: Encrypts victim data, asks payment to restore
Rootkits: SW hides malicious acts, maintains priv. access

DENIAL OF SERVICE

Overloading one or multiple systems to make it unavailable
SYN-Flood Attack: Sending many TCP connection requests without completing them
Reflection Attack: Attacker spoofs victim's IP, sends requests to a service so that it sends (many) replies to victim

WEB APPLICATION ATTACKS

Exploiting vulnerabilities in websites/servers hosting websites
Cross-Site Scripting (XSS): Inject malicious scripts into website that execute in users' browsers to steal data
Cross-Site Request Forgery (CSRF): Trick logged-in user's browser into sending reqs to a webapp on attacker's behalf
Broken Authentication: Weak auth mechanisms allow attackers to gain unauthorized access (password/session)

PASSWORD / AUTHENTICATION ATTACKS

Compromising login systems to gain unauthorized access
Rainbow Table Attacks: Precomputed hash lookup tables to reverse weakly hashed passwords back into plaintext
Password Spraying: Trying a few common passwords like "password" across many accounts to avoid lock-/timeouts
Credential Stuffing: Using leaked credentials from previous breaches to attempt logins on other services
Brute Force Attack: Repeatedly try many username & password combinations until they gain access to an account

PHYSICAL THREATS

Bypass technical cntrols by accessing physical infrastructure
Theft of devices: Physically steal hardware to gain direct access to stored data, credential, internal systems, …
Hardware tampering: Modify/implant malicious components in to intercept data, bypass security, or disrupt operations
Power disruption: Interrupt or manipulate power supply impacting availability and business continuity
Environmental damage: Environmental events that damage infrastructure, causing data loss, downtime, …

INFORMATION SECURITY MANAGEMENT (ISM)

Information security governance: System by which IS strategy is controlled to ensure that it supports business objectives, manages risk appropriately, and complies with legal and other regulatory requirements. (**what**)
Information Security Management System (ISMS): Framework used to manage and protect assets through policies, processes and controls (**how**)

Enterprise Information Security Policy (EISP): High-level Information security policy that sets the strategic direction and scope for all an organization's security efforts
Statement of Purpose: Scope, objectives, and purpose
Info Security Elements: Core principles & concepts (CIA)
Need for IS: IS importance & legal/ethical responsibility
IS Responsibilities and Roles: Organizational structure
Risk Management Process: Definition of processes to identify assets, analyze threats and evaluate risk
Security Awareness and Training: Educational programs to ensure employees know their security responsibilities
Monitoring, Measurement and Audits: Ongoing evaluation of control effectiveness and ISMS performance
Issue-specific security policy: Detailed, targeted guidance to instruct people in the use of a specific resource

POLICY

Practices: Ex. actions that illustrate compliance with policies
Policy: Instructions that dictate certain behavior within an org
Standard: Details of what must be done to comply with policy
Guidelines: Non-mandatory recommendations (reference)
Procedures: Step-by-step instructions to assist employees
De jure standard: Formally evaluated and approved by org
De facto standard: Widely adopted/accepted by public group
What does a policy do?: Establishes authority, account., responsib. Foundation for standards, procedures, guidelines
Who is responsible for policies?: Created by senior mgmt. Enforced by mgmt. Employees responsible for compliance
How is it enforced?: Communicate, integrate into standards, monitor compliance, disciplinary measure on violation
Cyber Resilience Act (EU): Requires secure-by-design digital products and vulnerability management

DESIGNING EFFECTIVE POLICIES

1) Development: Must align with organizational goals, business risks and legal requirements
2) Distribution: to all affected entities in a timely manner
3) Comprehension: Readable for, available to and read by all
4) Compliance: Formally agreed to by act or affirmation
5) Enforcement: Uniformly applied to all affected entities
6) Review: Reviewed regularly in a changing environment

RISK ANALYSIS

Identify Assets → Identify Threats → Identify Vulnerabilities → Assess Likelihood → Assess Impact → Determine Risk Level
Asset: **Item of value belonging to an organization. eg:**
Information: Customer data, intellectual property, code
Technical: Services, applications, databases, networks
Physical: Servers, devices, facilities, infrastructure
Human: Employees, administrators, contractors
Business Process: Critical operational workflows
Threat: An event or action with the potential to cause harm by exploiting a vulnerability. eg: power outage, vishing

CLASSIFYING ASSETS

Public: Information that can be shared without risk
Internal: Information for organization internal use only
Confidential: Sensitive info, could cause harm if disclosed
Restricted: Highly sensitive,strictly limited,strongly protected

SECURITY CONTROLS

Administrative/Management Controls: Policies, procedures, security training, security governance
Technical / Logical Controls: Firewalls, encryption, access control systems, system hardening
Physical Controls: Physical locks, surveillance cameras, secure access badges, turnstiles
Preventive Controls: Stop incidents before they occur. eg. Firewalls, access control, encryption
Detective Controls: Identify incidents when they occur. eg. Intrusion detection, log monitoring, SIEM, CCTV
Corrective Controls: Limit damage, restore systems after an incident. eg. Backups, system restore, incident response
Deterrent Controls: Discourage malicious behavior. eg. Warning banners, monitoring notices, disciplinary policies
Compensating Controls: Reduce risk when a primary control cannot be implemented. eg. Network isolation, layered security, alternative safeguards

Business continuity: Ensures critical operations continue during disruptions. Objectives: maintain operations, minimize impact, protect assets, recover fast

SECURITY AND AWARENESS TRAINING

1) **Awareness**: basic information, **what**
2) **Training**: detailed knowledge, **how**
3) **Education**: depth of knowledge, **why**

GAP ANALYSIS

Define target framework → Assess curr state → Identify gaps, assess risk → Eval, prioritize gaps → Create remediation plan

SECURITY FRAMEWORK

NIST Cybersecurity Framework: Framework for managing cyber risk: Identify, Protect, Detect, Respond, Recover
CIS Controls: Defines 18 practical technical security controls

THREATS

Attack: Act that intends to damage, steal or degrade assets
Vulnerability: A weakness in a system that can be abused
Exploit: A method used to take advantage of a vulnerability
Risk: The likelihood of a threat exploiting a vulnerability and the potential harm that could cause = Vulnerability + Threat
Threat vector: Path, method, or delivery mechanism that a threat uses to reach an asset and exploit a vulnerability
Attack surface: ∑ of threat vectors hackers can use to attack

RISK MANAGEMENT (RM)

The process of identifying, assessing, prioritizing and mitigating threats to an asset from an organisation
Risk management process: Implementation, analysis, evaluation of the risk management framework (**doing**)
Risk assessment: Identification, analysis, and evaluation of risk as initial parts of risk management
Risk treatment & Risk Owner: Application of safeguards or controls to reduce the risks to an acceptable level
1) **Risk identification**: Where and what is the risk?
2) **Risk analysis**: How severe is the current level of risk?
3) **Risk evaluation**: Is the current level risk acceptable?
4) **Risk treatment**: How do I bring risk to acceptable level?

Risk management framework: Structure of the strategic planning and design of risk management efforts (**planning**)
1) **Exec Governance & Support**: Support by mgmt & usrs
2) **FW Design**: Define methods,risk appetite strategy
3) **Framework Implementation**: Rollout of the plan
4) **Monitoring & Review**: How effective is entire system?
5) **Continuous Improvement**: Adaption to new threats

Risk appetite (strategic): The quantity of risk that organizations are willing to accept, to achieve their goals
Risk tolerance (specific): The acceptable risk organizations are willing to accept for a specific asset
Residual risk: Remaining risk after controls were applied
Common Vulnerabilities and Exposures (CVE): Standard identification number for vulnerabilities
Common Vulnerability Scoring System (CVSS): Severity scores for vulnerabilities based on CIA (14d 2b remediated)

QUANTITATIVE RISK ANALYSIS

1) Assign Asset Value (AV)
   1) Identify the organization's information assets.
   2) Classify them.
   3) Categorize them into useful groups.
   4) Prioritize them by overall importance.
2) Calculate Exposure Factor (EF = Damage/AV) percentage of loss that an organization would experience if a specific asset is violated by a realized risk
3) Calculate single loss expectancy (SLE = AV × EF) Exact amount of loss if an asset were harmed by a threat
4) Assess the annualized rate of occurrence (ARO) Expected occurrence frequency of a threat within a year
5) Derive the annualized loss expectancy (ALE=SLE × ARO) possible yearly cost of all instances of a realized threat against an asset
6) Perform cost/benfit analysis of countermeasures

Safeguard: Reduce the ARO and/or reduce the SLE
Annual cost of a safeguard (ACS): $ / year
ALE with safeguards (ALE2): ALE with updated ARO/SLE
Value of a safeguard: ALE1 - ALE2 - ACS. Negative = bad
Risk Evaluation: Compare risk with risk appetite of the org

Important Indicators (Business Impact): Maximum Tolerable Downtime (MTD) Recovery Point Objective (RPO) Recovery Time Objective (RTO) Work Recovery Time (WRT)
Risk treatment:
Mitigation: Apply safeguards to eliminate remaining risk (Firewall, Training)
Transfer: Shift risks to other areas/entities (Outsourcing)
Acceptance: Leave assets vulnerability facing the current risk level (after formal evaluation)
Termination: Remove asset from the environment
Risk mitigation:
– Fix vulnerabilities
– Applying controls (tools, processes, rules to mitigate risk)
– Reduce final impact (If vulnerabilities happen)
Endpoint Detection and Response (EDR): Software watching for sus behaviour, responds with measures
Extended Detection and Response (XDR): Like EDR but watching everywhere (not just on endpoints)

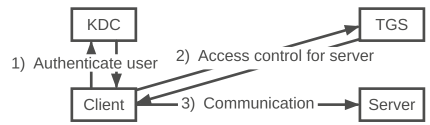
IDENTITY & ACCESS MANAGEMENT (IAM)

Provisioning and protecting digital ids & access permissions
Subject: Active entity that accesses a passive object. users, programs, processes, services, computers
Object: Passive entity that provides information to subjects. files, databases, computers, programs, services, printers
Physical controls: To prevent, monitor, or detect direct contact with systems: guards, fences, motion detectors
Technical or logical controls: To manage access and provide protection for resources. auth, encryption, ACL
Administrative controls: Policies and procedures. hiring practices, background checks, security training
Identification: Claiming an identity, eg. typing a username
Authentication: Verifying validity of claimed id, eg. password
1) Something you know, eg. password
2) Something you have, eg. smartcard
3) Something you are/something you do, eg. biometrics
4) Somewhere you are / aren't (secondary factor)

Basic Authentication: Username/pw transmitted in the clear
One Time Passwords: Basic auth but used only once
Challenge / Response: Password and one-time challenge
Anonymous Key Exchange: Exchange credentials over unauthenticated secure channel, eg. diffie-hellmann
Zero-Knowledge Password Proofs: Does not permit offline-based password attacks
Server Certificates + User Authentication: Transmit user password over unilaterally authenticated secure channel
Mutual Public Key Authentication: Bilateral use of public key signatures

KERBEROS

Uses symmetric key encryption (DES)



Principal: A Kerberos participant
Principal's Master Key: Long-term secret between the principal and KDC. Used to en-/decrypt authentication tickets.
Kerberos Ticket: Temporary credential for network services
Authentication Server (AS): Verifies client, gives TGT
Ticket Granting Server (TGS): Grants services, gives ST
Ticket Granting Ticket (TGT): Given by the AS
Service Ticket (ST): Given by the TGS
Key Distribution Center (KDC): Verifies & manages auth credentials, distributes session keys to users and services

RADIUS

Remote Authentication Dial-In User Service provides centralized Authentication, Authorization and Accounting (AAA)
1) User requests access from Network Access Server (NAS)
2) NAS prompts the RADIUS server for credentials
3) RADIUS server evaluates the request and returns one of:
   – Access-Accept: User authenticated, NAS grants access
   – Access-Reject: Invalid credentials, NAS denies access
   – Access-Challenge: Server requires more info (MFA)
4) Connected: NAS sends Accounting-Request (log session)

Attack	*Kerberos*	*RADIUS*
Passive Password Sniffing		
Offline Brute Force Password Attack	✗	✗
Active Man-in-the-Middle Attack		✗
Identity Theft on Server		
CA Compromise		✗

AUTHORIZATION

Access Control — Discretionary / Nondiscretionary
Nondiscretionary — Lattice-based — Mandatory / Role-/Task-based

Discretionary access control (DAC): Every object has an owner, the owner can grant or deny access (sharing drive)
Nondiscretionary access control (NDAC): Access controls that are implemented by a central authority (HIPAA/DSG)
Lattice-based access control (LBAC): Assigns users a matrix of authorizations for particular areas
Mandatory access control (MAC): Labels applied to both subjects and objects (classified documents ↔ glowies)
Role-based access control (RBAC): Privileges are tied to a role or job (project info ↔ project manager)
Task-based (TBAC) access control: Privileges are tied to a task (access to server room ↔ technician's time slot)
Auditing: A subject's actions are tracked and recorded
Accounting: Consumption of resources by a subject is measured, metered, and collected.
Access Aggregation Attacks (passive): Aggregate non-sensitive info to learn sensitive info (Reconnaissance attck)
Password Attacks (brute-force): Online (accounts) vs Offline (steal account database and crack the passwords)
Dictionary Attack (brute-force): Discover passwords by using every possible password in a predefined database
Birthday Attack (brute-force): Finding collisions. MD5 is not collision free, SHA-3 is safe against collisions
Sniffer Attacks: Application that captures traffic traveling over the network. Encrypt all sensitive data, Use OTP.

CRYPTOGRAPHY

Objectives: Confidentiality, Integrity, Auth, Non-repudiation
Steganography: Embedding secret messages within content
Nonce: Unique number for each usage. Makes sure that key is not re-used. Nonce is public, (shared) key is private
One-Way Functions: Easily produces output values but makes it impossible to retrieve the input values
Reversability: Being able to undo the operation of encryption
Ciphertext/Cryptogram: Encrypted message
Cipher: Encryption algorithm. Set of rules for en-/deciphering
Key/Cryptovariable: Usually a very large binary number
Key space: Range of numbers from 0 to 2^n, where n is the bit size of the key. A 128-bit key is $\in \{0, \dots, 2^{128}\}$
Initialization vector (IV): Random bit string, same length as the block size and is XORed with the message to create unique ciphertext every time same message is encrypted
Kerkhoff's Principle: Security stems from the secrecy of the key and not the secrecy of the algorithm
Shannon's Principles:
Confusion: Relationship between the plaintext and the key is complicated. S-Box, substitution
Diffusion: Change in plaintext results in multiple changes spread throughout the ciphertext. P-Box, permutation
SP-Network: Repeated substitution and permutation
One-Time-Pad (OTP): XORing all bytes with same size OTP
Hashing: Maps data to fixed-size output. Should be: random, diffused, fast, deterministic, irreversible, collisionless
Password storage: Prefer slow hash functions (SHA-256)
Summarizing data: Prefer fast hash functions (argon2)
HMAC: Hash based MAC, splits a key in two and hashes twice, not vulnerable to length extension attack
SHA-1: Insecure, fast
SHA-2: 256/512-bit variants, current standard, secure
SHA-3: (KECCAK), flexible, ≈ as secure as SHA-2
KMAC 128/256: SHA-3 based KECCAK MAC
bcrypt: Salted, GPU-resistant
Argon2: Highly secure, configurable
AES: Advanced Encryption Standard (SP-Net). XOR, SubBytes, ShiftRows, MixColumns, repeat (128 bit block size)

SYMMETRIC CRYPTOGRAPHY

Relies on shared secret key, distributed to all members before communication. No non-repudiation or message integrity.

Stream ciphers: Approximate OTP by generating infinite random keystream, work on messages of any length, nonce guarantees uniqueness, fast, no guaranteed integrity

Block ciphers: Take input of fixed size and return output of same size, use confusion and diffusion (SP-Network) **AES**

Electronic Code Block (ECB): Encrypt blocks sequentially with same key. Weak to redundant data divulging patterns

Cipher Block Chaining (CBC): XOR the IV with first input, then XOR output with next input. Not parallelizable

Counter Mode (CTR): Encrypt counter to produce stream cipher and XOR it with message. Parallelizable. **AES**

ASYMMETRIC CRYPTOGRAPHY

Relies on mathematically linked key pairs. $n \in \mathbb{N} \setminus \{0\}, z \in \mathbb{Z}$

Public-key enc: pubk (e, n) **encrypt**, privk (d) **decrypt**

Digital signing: privk (d) **sign**, pubk (e, n) **verify**

Discrete logarithm: $f(g, a, p) = g^a \bmod p$ (Diffie-Hellman)

Primitive root: g is primitive root of p if $g \bmod p, g^2 \bmod p, \dots, g^n \bmod p$ are distinct

Integer Factorization: $f(p, q) = p \cdot q$ (RSA)

Euler's Theorem: $\gcd(z, n) = 1 \Rightarrow z^{\varphi(n)} \equiv 1 \bmod n$

Totient:

$$\mathbb{Z}_n^* = \{x \in \mathbb{Z}_n \mid x \text{ has a multiplicative inverse in } \mathbb{Z}_n\}$$

$$\begin{aligned}\varphi(n) &= \text{Nr. of elements in } \mathbb{Z}_n \text{ with mult. inv.}\\ &= \text{Amount of Numbers } 1 \le q \le n \text{ with } \gcd(q, n) = 1\\ &= |\mathbb{Z}_n^*|\end{aligned}$$

1) $n \in \mathbb{N}$ prime $\Rightarrow \varphi(n) = n - 1$
2) $n \in \mathbb{N}$ prime, $p \in \mathbb{N} \setminus \{0\} \Rightarrow \varphi(n^p) = n^{p-1} \cdot (n-1)$
3) $m, n \in \mathbb{N} \setminus \{0\}, \gcd(m, n) = 1 \Rightarrow \varphi(n \cdot m) = \varphi(n) \cdot \varphi(m)$

RSA: Rivest–Shamir–Adleman: Signing, use DH for Encryption. Weak for short messages, add OAEP

1) Choose two prime numbers p, q
2) Calculate $n = p \cdot q$
3) Calculate $\varphi(n) = (p-1)(q-1)$
4) Choose a, b, so that $a \cdot b \equiv 1 \bmod \varphi(n)$
5) Forget $p, q, \varphi(p \cdot q)$

Public key is now n, b, private key is n, a

- Encrypt: $z = c^a \bmod n$
- Decrypt: $c = z^b \bmod n = c^{ab} \bmod n$

Optimal Assymetric Encryption Padding (OAEP): Introduces IV and hashes it `[0x00 | maskedSalt | maskedDB]`

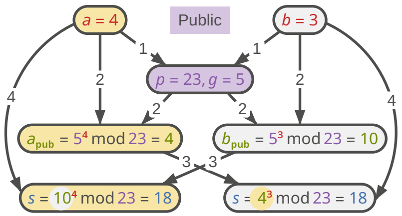
Probabilistic Signature Scheme(PSS): Hash,Salt,Pad,RSA

DSA: Digital Signature Algorithm

DSS: Digital Signature Standard

DH: Diffie Hellman, share secret over insecure channel

1) Agree on **public parameters** (prime and generator)
2) Combine **private key** with **the parameters**
3) Send resulting **public keys** to each other
4) Combine other **pubkey** with **priv key** to get shared **secret**



ECC: Elliptic Curve Cryptography $y^2 = x^3 + ax + b$

ECDSA: Elliptic Curve Digital Signature Algorithm

ECDSH: Elliptic-Curve Diffie Hellman

ECDHE: Elliptic Curve Diffie-Hellmann Ephemeral

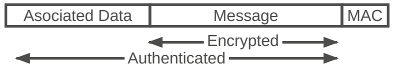
Perfect forward secrecy: Ephemeral DH KEX, forces new key exchange for every new session (refreshing website)

TRANSPORT LAYER SECURITY (TLS)

TLS < 1.2 insecure, 1.2 if configured correctly, 1.3 secure

TODO:
handshake

AEAD: Authenticated Encryption with associated data



GCM: Galois Counter Mode → Message Authentication Code

PUBLIC KEY INFRASTRUCTURE (PKI)

A set of roles, policies, hardware and software needed to manage digital certificates and public-key encryption

TODO:
explanations, cert fields

Subscriber $\xrightarrow{\text{CSR}}$ RA → CA → VA

RA: Registration Authority, validates CSR

CP: Certificate Policies

CPS: Certificate Practice Statements

CA: Certificate Authority, trusted certificate issuer

- **CSR**: Certificate Signing Request (creates)
- **CRL**: Certificate Revocation List (signs)

VA: Validation Authority, validates integrity of certificates

TSA: Time Stamp Authority

Subscriber: Client / Certificate holder

Certificate pinning (HPKP): Explicitly trusts only one certificate, all other root anchors are ignored. Used to prevent MITM. 3 different models: root, intermediate CA, end entity pinning. Poses big risk, because certificates are no longer fully validated. Legacy since 2017.

Truststore: Central (private) storage for certificates

Keystore: Central (private) storage for private keys

Hybrid cryptosystem: Var. methods for optimal use-cases Hashing: Digital fingerprint, Symmetric crypto: Bulk encryption, Asymmetric crypto: Signing/exchanging keys

Cipher suites: enc_kex_signature_WITH_bulk-enc_mac TLS_DHE_RSA_WITH_AES_128_GCM_SHA256

OCSP: Certificate Status Protocol

CT: Certificate Transparency

TODO:
illustrate CSR procedure

E-MAIL

Multipurpose Internet Mail Extensions (MIME): Contains various header fields and is split into multiple body parts

Secure MIME (S/MIME): Provides signatures + encryption. Central auth. sign-then-encrypt

Encapsulated S/MIME: Content within CMS SignedData Obj. Protected agains modifications

Encrypted S/MIME: Content within CMS EnvelopedData Obj. Protected against transfer encoding changes

Pretty Good Privacy (PGP): Uses Key pairs, no central auth (direct trust/indirect trust). encrypt-then-sign

Sign-then-encrypt: signer knows message, signature hidden, oracle attacks, no integrity, recipient could re-encrypt

Encrypt-then-sign: signature ensures integrity, signer might not know message, recipient could re-sign

Sender Policy Framework (SPF): Specifies which servers may send email for domain (DNS TXT: SMTP MAIL FROM)

DomainKeys Identified Mail (DKIM): PKI based on DNS

Domain-based Message Authentication, Reporting, and Conformance (DMARC): Combines SPF + DKIM

WEB

Common Weakness Enumerations (CWE): Classification of types of vulnerabilities (eg. XSS, SQL Injection)

Open Web Application Security Project (OWASP): Community-driven project for web application security

Broken Access Control: Eg. get data by changing URL ID

- **Avoidance Insecure Direct Object References (IDOR)**: prevents this (eg. UUIDs, server-side auth checks)
- **Centralized Access Control Logic**: Handle in backend

Injection: SQL, Cross-site Scripting (XSS), Shell

- **(1) Secure Programming**: Prepared statements
- **(2) Infrastructure security**: DB Least privileges, WAF
- **(3) Secure Programming**: No logs 4 user, Anon error msgs

Reflected XSS: Data provided by web client is used immediately by server to generate a page of results for that user

Stored XSS: Data provided by web client is stored in a database. This data is then presented to the user unencoded

DOM-based XSS: Original client side JS embeds attacker's payload (eg. from URL) into DOM → `eval()` or `innerHTML`

- **(1) Secure Programming**: Input sanitization
- **(2) Security Mechanisms**: CSP, Headers, WAF

CYBER KILL CHAIN

A model developed by Lockheed Martin in 2011

1) Reconnaissance: Gather information on the target
   - Social media, DNS Lookup, Legal info
2) Weaponization: Construct custom weapon to attack target
   - Exploit: Office macro execution, malicious PDF
   - Payload: Remote Access Tool, Ransomware, Spyware
   - Weapon: Phishing Email, Crafted Network Request
3) Delivery: Transmit the weapon to the target
   - USB Drops, drive-by downloads, phishing mails
4) Exploitation: Vulnerability is triggered or the target tricked
5) Installation: Establish persistence, install backdoors
   - Scheduled tasks, startup folders
6) Command & Control (C2): Compromised host connects to the attacker
   - DNS tunneling, HTTPS to legitimate-looking domain
7) Actions on Objectives: Exfiltration, encryption, sabotage, lateral movement

Beaconing: Most C2 traffic beacons: small periodic check-ins, often jittered, tiny in volume but constant in rhythm

Defense-in-depth mapping: Each control (EDR, IDS, backup) maps to phases. Gaps = phases without coverage

Dwell time matters: Median dwell time = 10 days for an attacker to move between phases, undetected

Metric you can measure: Time from phase-1 artefact entering your environment to phase-7 impact

Detections, not tools: Buying appliance without knowing which phase it improves coverage on is expensive theatre

MITRE ATT&CK: Matrix of 14 tactics, hundreds of techniques

INCIDENT RESPONSE

ESCALATION PATH

Event: *Any observable occurrence in a system. A user logs in, a file is created, a packet arrives. Most events are routine*

Alert: *Automated notification from a security tool that an event matched a detection rule and may warrant investigation*

False positive: *An alert or suspected adverse event that turns out to be benign*

Adverse event: An event with possibly negative consequences, worth investigating

Incident: A confirmed adverse event that threatens the confidentiality, integrity, or availability of information assets

Data breach: An incident resulting in unauthorised access to or disclosure of protected data

PIPKIN'S INDICATORS

Possible: Weak signals, worth logging but not acting on alone

Probable: Strong signals, typically trigger investigation

Definite: Confirmed incidents, activate the IR plan

Classifier's dilemma:

True Positive	False Positive
False Negative	True Negative

DETECT AND RESPOND

"Prevent Everything": If the perimeter holds, we are safe

"Assume Breach" Mindset: Design for fast detection and fast containment, not just for prevention.

MTTD (Mean Time to Detect): Initial compromise - first alert

MTTR (Mean Time to Resp.): First alert - incident contained

Dwell Time (MTTD + MTTR): Total time attacker was active

SOC (Security Operations Centre): The always-on monitoring function. Runs the SIEM, triages alerts, drives day-to-day detection. (T1 triage, T2 investigation, T3 engineering).

CSIRT (Computer Security Incident Response Team): Handles confirmed incidents end-to-end. Often virtual, pulled together when needed: IT, legal, comm, mgmt

CERT (Computer Emergency Response Team): National or sector-level coordination bodies (e.g., GovCERT.ch)

In-House vs. Managed: Many organisations outsource SOC functions to an MSSP (Managed Security Service Provider). CSIRT responsibilities usually stay in-house.

NIST IR PROCESS

1) Preparation: Policy, plan, team, tools, training
2) Detection & Analysis: Spot signals, classify them
3) Containment, Eradication & Recovery: Stop the bleeding, remove the attacker, restore services. Short/Long-Term
4) Post-Incident Activity: Lessons learned, feed back to Prep

COMMON IR MISTAKES

Unclear Chain of Command: Attacker uses the confusion

No Central operations center: Decisions are undocumented

Containing too slowly: Fear of "making it worse"

Wiping evidence b4 Forensics: Rebuilding infected server

Skipping after-action Review: Same mistakes next year

No tested Backups: Everyone assumes the backups work

INTRUSION DETECTION SYSTEMS (IDS)

IDS: Monitors systems, flags activity, raises alerts

Intrusion Prevention System (IPS): IDS that takes action

Signature-based detection: Match traffic w/ known pattern

Anomaly-based detection: Alert on deviations from "normal"

Stateful Protocol Analysis: IDS understands net protocols

Heuristic & Behavioral Rules: Rules describing sus seq.

AI-Assisted Classification: ML trained on labelled samples

Hybrid in Practice: Real-world products blend all the above

SECURITY INFORMATION AND EVENT MANAGEMENT

1) Sources: Endpoints, firewalls, servers, identity providers
2) Collectors: Agents, syslog receivers, API, log forwarders
3) Parse & Normalize: Map every log into a common schema
4) Correlate & Store: Detection rules, ML, search index
5) Consume: Alerts to the SOC, dashboards, search, reports

TODO:
SIEM correlation, SOC

SIEM: Collect, normalize, correlate, alert. Human response.

Security Orchestration, Automation and Response (SOAR): On top of SIEM. Runs playbooks: isolate, disable

Extended Detection and Response (XDR): Vendor-bundled detection. Lighter alt. to SIEM+SOAR, vendor lock-in.

In real enterprises: SIEM + SOAR is the classic stack

Volume Problem: Avg company produces 20+GB logs/day

Correlation Problem: Attacks touch several systems

Retention Problem: Forensics need logs from 90+ days ago

Compliance Problem: PCI-DSS, ISO 27001, FINMA, NIS2

Alert Fatigue: Many low-quality alerts, analysts stop reading

Garbage In / Garbage Out: Incomplete, mis-parsed logs

Ingestion Cost: Most SIEMs price per GB per day

Tuning Debt: Rules that fit last year do not fit this year

Skills Gap: Detection engineers/threat hunters needed

OPEN-SOURCE INTELLIGENCE (OSINT)

Collecting, analyzing, decisions based on public information

Cyber Threat Intelligence (CTI): Proactive collection, processing, analysis of information about threats

External sources: CI vendors, Subscription service

Internal sources: Logs, Alerts, Dedicated teams

Advanced Persistent Threat (APT) Attack: Attacker gains access to network, stays there, undetected, for long time

Research Intel > Attack Exploit > Persist C2 > Move Priv. Esc. > Exfiltrate Data > Maintain Target notified

1) Strategic Level for Executives & Management: Who is attacking and why?
2) Operational Level for SOC Teams & Analysts: How does a specific attack unfold?
3) Tactical Level for SIEM systems & Firewalls: Which concrete indicators do I need to block?

ETHICAL HACKING

Validate, audit and report on system/software vulnerabilities

Black Hat: *Malicious, destructive hacker, anonymous*

Grey Hat: *Possess Black hat skills, focus on offense+defense*

White Hat: *Possess Black hat skills, focus on defense*

Script Kiddie: *Use tools without knowing what they are doing*

Cyber Terrorist: *Skilled attacker with ideological purpose*

State Sponsored: *Hackers employed by the government*

Hacktivist: *Hacking in order to pursue a political or social aim*

Pentesting: Manual process, in-depth analysis

Vulnerability scanning: Automated, periodic

Penetration Testing Execution Standard (PTES): Community-driven industry standard, e2e pentesting methodology

NIST 800-115: Technical Guide, Government-oriented

EC-Council: Global organization providing cysec certs

Black-box Testing: No internal info, Attacker's perspective

Gray-box Testing: Limited knowledge, Realistic auth, Insider-based threat modeling

White-box Testing: Full internal knowledge, Deep analysis

Statement of Work (SoW): Activities to be performed, timeline, location, scope. → Master Service Agreement (MSA)

Non Disclosure Agreement (NDA): Uni-/Bi-/Multilateral

MALICIOUS CODE

User-Dependent Malware: Most malware rely on user interaction or unsafe behavior to propagate

Self-Replication Threats: (Worms) – no human involvement

Computer Viruses: Propagation and payload execution

Drive-by downloads: Unintentional dl of malicious code

MBR infection: Read MBR → Exec malware → Start OS

File infection: Infect exe files, self-contained, easily detected

Service injection: Inject into trusted runtime processes

Macro infection: Written in embedded macro lang (Excel)

Fileless techniques: RAM, resistant to signature-based det.

Multipartite Viruses: Use > than one propagation technique

Stealth Viruses: Hide themselves, trick antivirus

Polymorphic Viruses: Modify their code as they travel. Signature differs, antivirus vendors cracked the code of many

Encrypted Viruses: Use cryptographic techniques, include decryption routine segment, use different keys (polymorph)

Logic Bombs: Lie dormant until some event triggers them

Spy-/Adware: windows

ANTIVIRUS & ENDPOINT SECURITY

Signature-based: Database of characteristics of viruses

Heuristic-based: Analyze the behavior of software

Data integrity: Detect Unauthorized file modifications (hash)

Endpoint Security: EDR tools capture all sys events. Behavioral Analysis, Automated Response and Remediation

POST-QUANTUM CRYPTOGRAPHY

Quantum Computer: Factoring, Discrete Log, Hashing, Asymmetric Crypto (RSA), Symmetric Crypto (HMAC)

IOT

Feature	*IT (Office/IT)*	*IoT (Connected)*	*OT (Industrial)*
CIA Prio	C > I > A	I > C > A	A > I > C
Asset Focus	Servers, Laptops, Databases	Smart Dev, Trackers, Cams	Motors, Robots, Sensors
OS Type	Windows, Linux, MacOS	Embedded / Lightweight Linux	Real-time (RTOS) Proprietary
Lifecycle	3–5Y Short-lived	2–7Y Fast-paced	15–30Y Legacy S.
Patching	Regular (e.g. "Patch Tuesday")	OTA (Over-the-Air) targeted	Rare (only during downtime)
Failure Impact	Data loss, Reputation damage	Mass manipulation, Botnets	Physical damage, Risk to life

FUTURE

Security-by-Design, Zero-Trust Architectures, Automated patch mgmt & standardization, Compliance with standards

TODO:

- iso
  ISO 27000 is a certifiable international standard, where the NIST Cybersecurity Framework does not have certification.
- The Kerberos KDC stores a list of hashes of the principals' passwords. If an attacker steals this list, what is the most accurate consequence?
  - Because the hash is used directly as the symmetric key (Master Key), an attacker with the hash can impersonate the user without needing the plaintext password.
  - The password itself never traverses the network. The client uses the hash of the password as a symmetric key (the principal's Master Key) to encrypt/decrypt

    messages. This means only encrypted material cross the wire.
- Encrypting a message so only one specific recipient can read it. → Recipient’s public key, Decrypting a message that was sent to me. → My own private key, Producing a digital signature on a message. → Signer’s private key, Verifying a digital signature. → Signer’s public key
- TLS 1.3
  - Establishes a shared secret with perfect forward secrecy. → ECDHE (Ephemeral Elliptic Curve Diffie-Hellman), Proves the server’s identity (and optionally the client’s) via a digital signature. → RSA / ECDSA / EdDSA signatures, Derives traffic keys and IVs from the handshake secret. → HKDF, Provides confidentiality + integrity for application data in a single primitive (e.g., AES-GCM, ChaCha20-Poly1305). → AEAD cipher
- RSA vis

This summary was created thanks to the motivation provided by turnstile , wisdom in chains and hands of god